Radius of Curvature	Distance between the center of the sphere and the vertex.
Reflection	Bouncing back of the light rays as they fall on the smooth, shiny and polished surface.
Refraction of light	Bending of light about the normal, at the point of incidence; as it passes from one transparent medium to another.
Refractive index	Ratio of the speed of light in the air to the speed of light in that particular medium.

TEXTBOOK EXERCISES

I. Choose the best answer.

1. Which of the following has curved reflecting surface?
   a) plane mirrors b) spherical mirrors
   c) simple mirrors d) None of the above
2. The spherical mirror with a reflecting surface curved inward is called
   a) convex mirror b) concave mirror
   c) curved mirror d) None of the above
3. The spherical mirror used as a rear view mirror in the vehicle is
   a) concave mirror b) convex mirror
   c) plane mirror d) None of the above
4. The imaginary line passing through the centre of curvature and pole of a spherical mirror is called
   a) centre of curvature b) pole
   c) principal axis d) radius curvature
5. The distance from the pole to the focus is called
   a) pole length b) focal length
   c) principal axis d) None of the above
6. If the image and object distance is same, then the object is placed at
   a) infinity b) at F
   c) between f and P d) at C
7. If the focal length of a spherical mirror is 10 cm, what is the value of its radius of curvature?
   a) 10 cm b) 5 cm c) 20 cm d) 15 cm

II. Fill in the blanks.

1. The spherical mirror used in a beauty parlour as make-up mirror is _____.
2. Geometric centre of the spherical mirror is _____.
3. Nature of the images formed by a convex mirror is _____.
4. The mirror used by the ophthalmologist to examine the eye is _____.
5. If the angle of incidence is 45°, then the angle of reflection is _____.
6. If an object is placed between two mirrors which are parallel to each other, the number of images formed is _____.

III. Match the following.

Convex mirror	Radio telescopes
Parobolic mirror	Rear – view mirror
Snell's law	Kaleidoscope
Dispersion of light	sin i/sin r =μ
Refractive index	Rainbow

8th_Science_Unit-3_EM.indd 33 12/8/2022 3:02:18 PM